Bayesian Fraud Detection

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1 Introduction

Modern financial systems generate large volumes of transactional data that evolve over time. A key challenge in this context is the detection of anomalous or potentially fraudulent transactions.

This report addresses the problem of anomaly detection in financial transaction data from a Bayesian perspective.

While my personal thesis project develops a structured modeling framework based on Relational Event Models (REMs) and latent variables for considering anomalies, here we consider a different but complementary viewpoint.

Instead of explicitly modeling the interaction structure between entities and the temporal evolution of events, we directly model the probability that a transaction is anomalous. This leads to a different probabilistic formulation that focuses on inference rather than on the full generative dynamics of the system.

Within this setting, Bayesian methods provide a natural alternative to more complex event-based models. We first consider a conjugate Beta-Binomial model to estimate the global rate of anomalies, and then extend the analysis using a Bayesian logistic regression model estimated via MCMC to incorporate transaction-level features.

The first approach provides interpretability and analytical tractability, while the second allows incorporating transaction-level features.

The purpose of this work is to show how Bayesian inference can be used as an alternative, more compact approach to anomaly detection compared to REM-based latent variable models.

In this work, we address anomaly detection using a probabilistic perspective, framing the problem within a Bayesian framework.

2 Data Description

The dataset used in this work is publicly available on the Kaggle bank transaction dataset by Watsky and contains anonymous bank transaction records. Each observation corresponds to a single financial transaction over time, recorded in a banking system for a single account holder. Unlike relational transaction datasets, where interactions occur between a sender and a receiver, this dataset does not contain explicit dyadic structure. Instead, it represents a uni-variate sequence of financial events associated with one entity evolving over time, where each transaction affects the account balance.

The raw dataset includes variables such as transaction date, withdrawal amount, deposit amount, cheque number, and account balance. In the preprocessing phase, structurally irrelevant variables (e.g. cheque number and textual descriptions) were removed.

Since withdrawal and deposit represent mutually exclusive outcomes of the same transaction, they were merged into a single monetary variable to obtain a consistent representation of financial flow. Rows with missing or undefined numerical entries were excluded as part of a structural cleaning step, rather than generic imputation, to ensure coherence of the transaction sequence used in the analysis.

Variable	**Description**
amount	Unified transaction value derived from withdrawal/deposit
balance	Account balance after or before the transaction
date	Timestamp of the transaction
time index	Time from the first occurred event
log time	Log-transformed time index
is withdrawal	Binary indicator for withdrawal transactions
balance amount ratio	Ratio between balance and absolute transaction amount
weekday	Day of the week of the transaction
Y	Binary anomaly label obtained via Isolation Forest

Table 1: Summary of variables used in the analysis

After preprocessing, the dataset contains 116201 transactions.

3 Anomaly Label Construction

Since true fraud labels are not available, for the first part of the project we construct a proxy variable Y_i using an unsupervised anomaly detection model.

We employ Isolation Forest to assign anomaly scores to all transactions. In particular, a transaction is labeled as anomalous if its score exceeds the 95th percentile:

$$Y_i = \begin{cases} 1 & \text{if anomaly} \\ 0 & \text{otherwise} \end{cases}$$

This produces a binary indicator used in the Bayesian models. The Isolation Forest procedure identifies 5810 anomalies, corresponding to an anomaly rate of approximately 5.00%. The threshold used to define anomalous observations is 0.472158.

Quantity	Value
Total observations	116201
Detected anomalies	5810
Anomaly rate	0.0500
95% score threshold	0.4722
Minimum anomaly score	0.3529
Median anomaly score	0.3877
Mean anomaly score	0.3966
Maximum anomaly score	0.7756

Table 2: Isolation Forest anomaly label summary

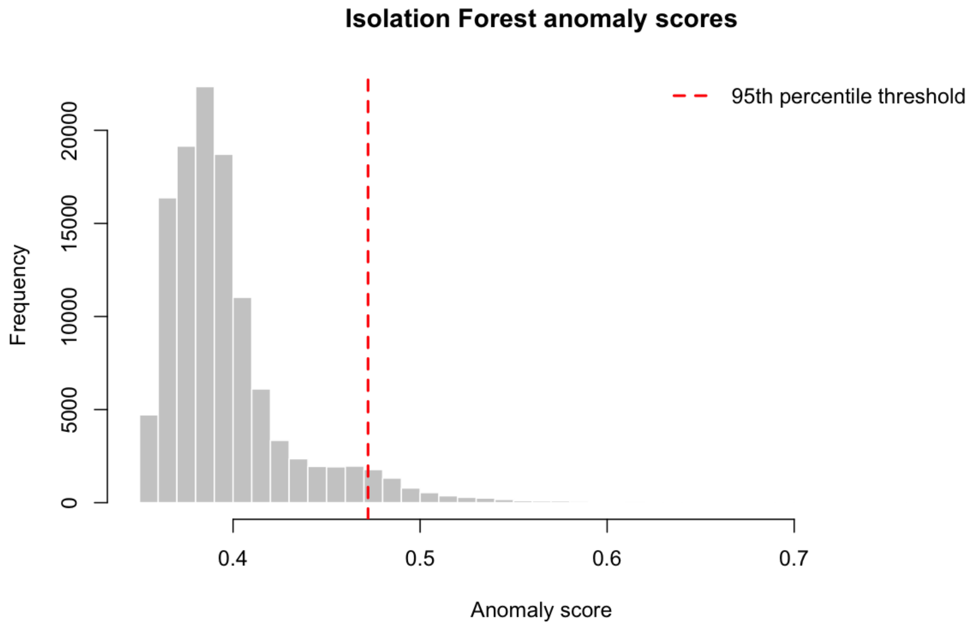

Figure 1: Distribution of Isolation Forest anomaly scores. The dashed line marks the 95th percentile threshold.

4 Bayesian Beta-Binomial Model

We now model the global anomaly rate π as $Y_i \sim \text{Bernoulli}(\pi)$, with prior parameter distribution $\pi \sim \text{Beta}(\alpha_0, \beta_0)$.

This choice is motivated by the conjugacy between the Beta prior and the Bernoulli likelihood. Conjugacy implies that the posterior distribution belongs to the same family as the prior, leading to a closed-form update:

$$\pi \mid Y \sim \text{Beta}(\alpha_0 + k, \beta_0 + n - k),$$

where $k = \sum_{i=1}^{n} Y_i$ is the number of observed anomalies of n observations.

4.1 Interpretation

The posterior mean is:

$$\mathbb{E}[p \mid Y] = \frac{\alpha}{\alpha + \beta}.$$

The Beta-Binomial analysis is performed on a random subsample of 250 transactions, in which 14 anomalies are observed. This corresponds to an empirical anomaly rate of 5.6% in the subsample.

Prior	α	β	**Posterior mean**	**95% CI low**	**95% CI high**
Beta(1,1)	15	237	0.0595	0.0338	0.0918
Beta(1,19)	15	255	0.0556	0.0315	0.0858
Beta(5,95)	19	331	0.0543	0.0331	0.0803

Table 3: Posterior estimates under different Beta priors

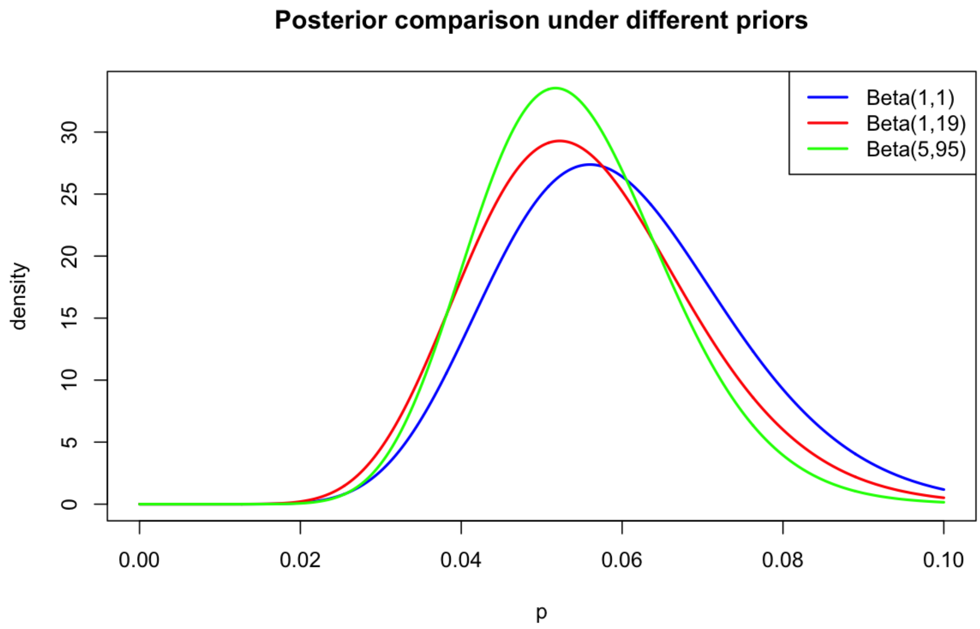

Figure 2: Posterior distributions of the anomaly probability under three different Beta priors.

5 Sequential Bayesian Learning

To analyze how the anomaly probability evolves over time, we consider an online update scheme.

At each time step t, we update:

$$\alpha_t = \alpha_0 + k_t, \quad \beta_t = \beta_0 + t - k_t,$$

where k_t is the cumulative number of anomalies observed up to time t.

The posterior expectation evolves as:

$$\mathbb{E}[p_t] = \frac{\alpha_t}{\alpha_t + \beta_t}.$$

This allows us to track how the estimated anomaly probability changes as more transactions are observed.

At the end of the sequential updating procedure, the posterior mean anomaly probability is 0.050007. The final 90% credible interval is $[0.048960, 0.051063]$, showing that after observing the full sequence the global anomaly rate is estimated very precisely.

Quantity	**Value**
Final posterior mean	0.050007
Final 90% credible interval lower bound	0.048960
Final 90% credible interval upper bound	0.051063
Minimum posterior mean over time	0.050007
Maximum posterior mean over time	0.984848

Table 4: Sequential Bayesian learning summary

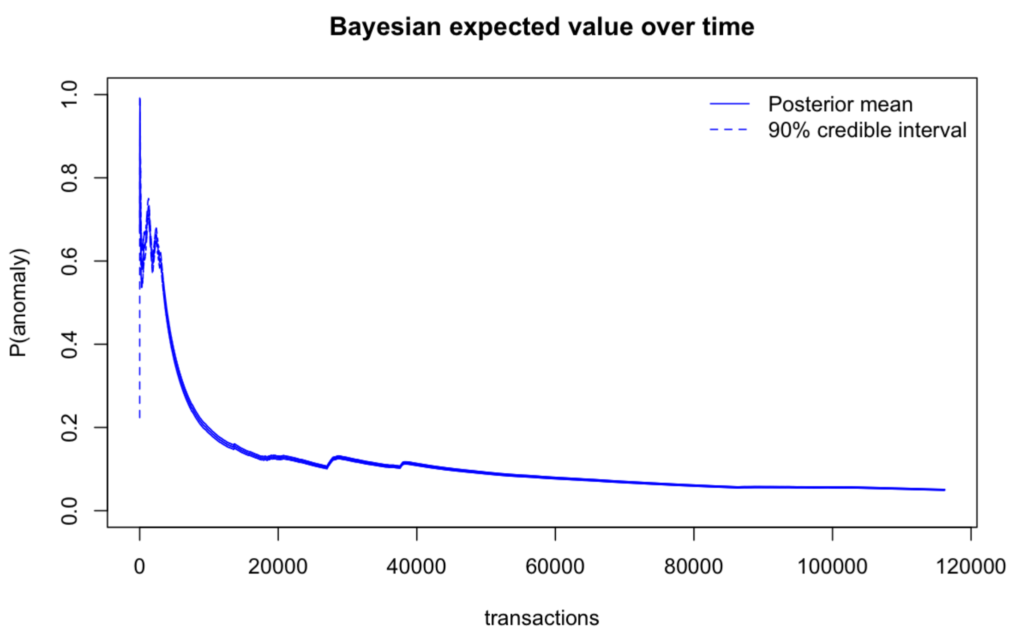

Figure 3: Sequential posterior mean of the anomaly probability with 90% credible interval.

6 Bayesian Logistic Regression via MCMC

While the Beta-Binomial model provides a global estimate of the anomaly rate, it does not incorporate transaction-level information. To model how transaction characteristics influence the probability of anomalous behaviour, we extend the analysis through Bayesian logistic regression.

Given a binary anomaly indicator $Y_i \in \{0, 1\}$ for transaction i, we assume $Y_i \sim \text{Bernoulli}(\pi_i)$ where π_i represents the probability that transaction i is anomalous. The probability is linked to the observed covariates through a logistic regression model:

$$\text{logit}(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4}.$$

The predictors include standardized transaction amount, standardized balance, standardized log-time, withdrawal indicator, balance-to-amount ratio, and weekday effects.

Differently from the Beta-Binomial setting, the posterior distribution of the regression coefficients cannot be derived analytically due to the non-conjugate logistic likelihood. For this reason, posterior inference is performed through Markov Chain Monte Carlo (MCMC) methods.

6.1 MCMC intuition

Instead of producing a single point estimate for the parameters, MCMC generates a sequence of samples:

$$\beta^{(1)}, \beta^{(2)}, \ldots, \beta^{(s)} \sim p(\beta \mid Y, X),$$

which approximate the posterior distribution of the model parameters. Each draw corresponds to one plausible configuration of the regression coefficients given the observed data.

More specifically, each sampled vector contains one coefficient for every feature included in the model:

$$\beta^{(s)} = \left(\beta_0^{(s)}, \beta_1^{(s)}, ..., \beta_n^{(s)}\right),$$

where each component represents the effect of a specific transaction feature on the anomaly probability.

The sampling procedure constructs a Markov chain whose stationary distribution coincides with the posterior distribution of the parameters. After an initial warm-up phase, the chain progressively explores the high-probability regions of the posterior space. The objective is to obtain samples that are as weakly correlated as possible, so that successive draws behave approximately as independent realizations from the target posterior distribution.

Predictions are therefore obtained by averaging over many sampled parameter configurations rather than relying on a single estimated model. This allows the Bayesian framework to naturally quantify predictive uncertainty.

In particular, for each transaction the model provides:

- a posterior mean probability of fraud;
- a credible interval describing uncertainty around the prediction.

This represents one of the main conceptual differences with respect to the frequentist logistic regression approach, which produces only point estimates of the probabilities.

The Bayesian logistic regression was fitted on 92960 training observations and evaluated on 23241 test observations. The training anomaly rate is 0.050054, while the test anomaly rate is 0.049783.

Quantity	**Value**
Post-warmup posterior draws	2400
Maximum $\hat{R}$ among fixed effects	1.002404
Minimum bulk ESS among fixed effects	1359.47
Minimum tail ESS among fixed effects	1222.73

Table 5: MCMC diagnostic summary

The convergence diagnostics are satisfactory: all fixed-effect $\hat{R}$ values are close to 1, and the effective sample sizes are sufficiently large for the fixed effects.

7 Comparison of Bayesian Approaches

The two Bayesian approaches considered in this work address anomaly detection at different levels of complexity.

The Beta-Binomial model provides a simple probabilistic description of the overall anomaly frequency in the dataset. Due to the conjugacy between the Beta prior and the Bernoulli likelihood, posterior inference admits a closed-form solution and can be computed analytically. This makes the model highly interpretable and computationally efficient, although limited to estimating a global anomaly probability.

The Bayesian logistic regression model extends the analysis by incorporating transaction-specific covariates. In this case, posterior inference no longer admits a closed-form expression and requires simulation-based methods such as MCMC. Although computationally more expensive, this approach allows the anomaly probability to vary across transactions according to their observed characteristics.

From a predictive perspective, the Bayesian logistic regression outperforms the frequentist logistic regression in ROC-AUC, log loss, and F1-score when both use a fixed threshold of 0.5. The optimal-threshold Bayesian classifier substantially increases recall, at the cost of lower precision and accuracy.

Model	Threshold	AUC	Accuracy	Log loss	Recall	F1
Frequentist logistic	0.5000	0.8399	0.9552	0.1429	0.1746	0.2794
Bayesian logistic	0.5000	0.8831	0.9614	0.1237	0.2774	0.4172
Bayesian logistic, optimal	0.1179	0.8831	0.9313	0.1237	0.7874	0.5329

Table 6: Predictive performance on the test set

Model	TP	TN	FP	FN
Frequentist logistic	202	21997	87	955
Bayesian logistic	321	22023	61	836
Bayesian logistic, optimal threshold	911	20733	1351	246

Table 7: Confusion matrix components on the test set

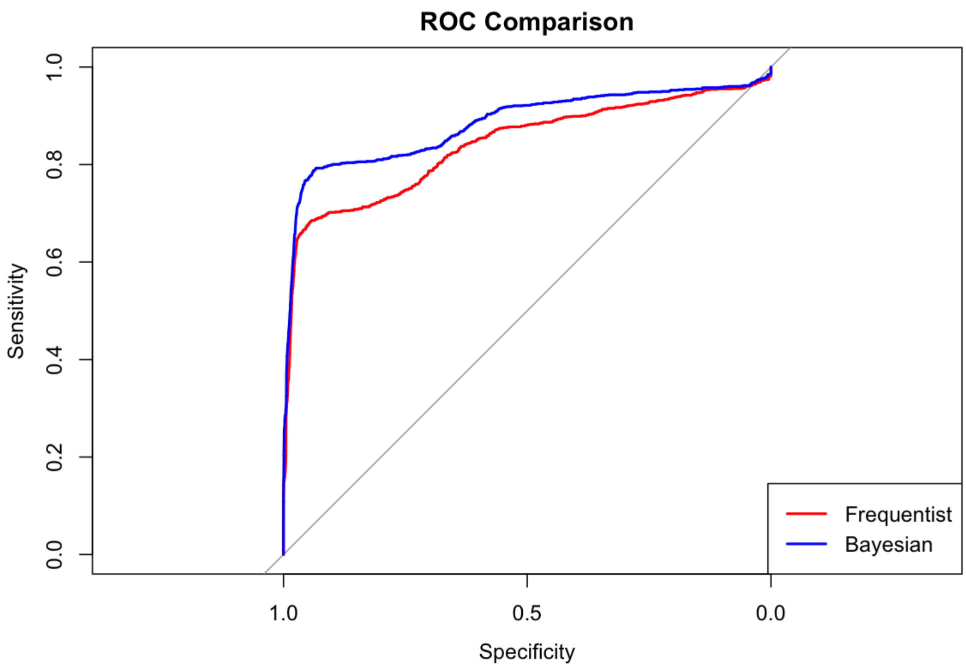

Figure 4: ROC curve comparison between frequentist and Bayesian logistic regression.

The posterior predictive probabilities have mean 0.050844 and median 0.009279. Their 5% and 95% quantiles are 0.000864 and 0.228405, respectively. The average width of the 90% credible interval is 0.006430, with median width 0.001762.

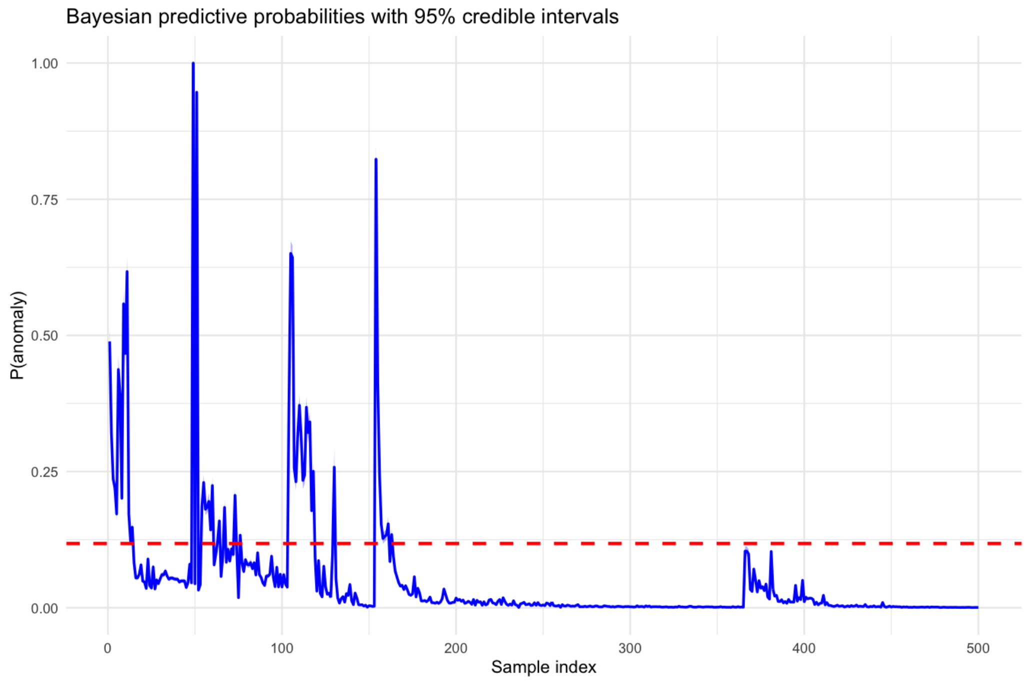

Figure 5: Bayesian posterior predictive probabilities with 90% credible intervals for a sample of test observations.

The Bayesian framework therefore provides not only point predictions and classification performance, but also a direct quantification of uncertainty. This additional uncertainty quantification is particularly relevant in fraud detection applications, where identifying uncertain or borderline cases may be as important as maximizing classification accuracy itself.